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Batch-51

ASSIGNMENT-16.2

Database Design and Queries: Schema Design and SQL Genera on
Task Descriptor-1: (Design Database Schema for Student Management

System) INPUT:

```
Schema SQL ●
1 CREATE TABLE Students (
2     student_id INT PRIMARY KEY,
3     name VARCHAR(100),
4     email VARCHAR(100)
5 );
6
7 CREATE TABLE Courses (
8     course_id INT PRIMARY KEY,
9     course_name VARCHAR(100),
10    credits INT
11 );
12
13 CREATE TABLE Enrollments (
14     enrollment_id INT PRIMARY KEY,
15     student_id INT,
16     course_id INT,
17     FOREIGN KEY (student_id) REFERENCES Students(student_id),
18     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
19 );
20
21 -- SHOW OUTPUT
22 SHOW TABLES;
23 DESCRIBE Students;
24 DESCRIBE Courses;
25 DESCRIBE Enrollments;
```

OUTPUT:

Results						Copy as Markdown
Query #5 Execution time: 0.33ms						
Field	Type	Null	Key	Default	Extra	
student_id	int(11)	NO	PRI	null		
name	varchar(100)	YES		null		
email	varchar(100)	YES		null		
Query #6 Execution time: 0.30ms						
Field	Type	Null	Key	Default	Extra	
course_id	int(11)	NO	PRI	null		
course_name	varchar(100)	YES		null		
credits	int(11)	YES		null		
Query #7 Execution time: 0.37ms						
Field	Type	Null	Key	Default	Extra	
enrollment_id	int(11)	NO	PRI	null		
student_id	int(11)	YES	MUL	null		
course_id	int(11)	YES	MUL	null		

Explan on

- 1. Defines structure of database using tables: Students, Courses, Enrollments.
- 2. Primary keys ensure unique identification of each record.
- 3. Foreign keys maintain referential integrity between related tables.
- 4. Follows normalization (1NF, 2NF, 3NF) to remove redundancy and ensure consistency.

Task Descriptor- 2(Insert Sample Records)

INPUT:

Database: MySQL v5.7 Run Update Fork Load Example Star PRO

Query SQL PRO Have any feedback?

```
1 INSERT INTO Students VALUES
2 (1, 'Arjun', 'arjun@gmail.com'),
3 (2, 'Meera', 'meera@gmail.com'),
4 (3, 'Rahul', 'rahul@gmail.com');
5
6 INSERT INTO Courses VALUES
7 (101, 'Database Systems', 4),
8 (102, 'Data Structures', 3),
9 (103, 'Operating Systems', 4),
10 (104, 'AI Fundamentals', 3);
11
12 INSERT INTO Enrollments VALUES
13 (1, 1, 101),
14 (2, 1, 102),
15 (3, 2, 103),
16 (4, 3, 101),
17 (5, 3, 104);
18
19 -- SHOW RESULT
20 SELECT * FROM Students;
21 SELECT * FROM Courses;
22 SELECT * FROM Enrollments;
```

OUTPUT:

Results Copy as Markdown

student_id	name	email
1	Arjun	arjun@gmail.com
2	Meera	meera@gmail.com
3	Rahul	rahul@gmail.com

Query #5 Execution time: 0.07ms

course_id	course_name	credits
101	Database Systems	4
102	Data Structures	3
103	Operating Systems	4
104	AI Fundamentals	3

Query #6 Execution time: 0.06ms

enrollment_id	student_id	course_id
1	1	101
2	1	102
3	2	103
4	3	101
5	3	104

Explain on:

1. Populates tables with initial data using INSERT statements.
2. Ensures relational links by matching student_id and course_id.
3. Helps in testing database functionality with sample dataset.
4. SELECT queries verify whether data insertion is successful.

Task Descriptor- 3: (CRUD Operations) INPUT:

Database: MySQL v5.7 Run Update Fork Load Example Star PRO

Query SQL PRO Have any feedback?

```

1 -- READ
2 SELECT * FROM Courses;
3
4 -- UPDATE
5 UPDATE Students
6 SET email = 'arjun_new@gmail.com'
7 WHERE student_id = 1;
8
9 -- DELETE
10 DELETE FROM Enrollments
11 WHERE enrollment_id = 5;
12
13 -- ☒ SHOW RESULT
14 SELECT * FROM Students; -- shows updated email
15 SELECT * FROM Enrollments; -- shows deleted record

```

OUTPUT:

Results	Copy as Markdown	
Query #1 Execution time: 0.13ms		
course_id	course_name	credits
There are no results to be displayed.		
Query #2 Execution time: 0.13ms		
There are no results to be displayed.		
Query #3 Execution time: 0.07ms		
There are no results to be displayed.		
Query #4 Execution time: 0.06ms		
student_id	name	email
There are no results to be displayed.		
Query #5 Execution time: 0.06ms		
enrollment_id	student_id	course_id
There are no results to be displayed.		

Explana on:

1. SELECT retrieves data from tables for viewing records.
2. UPDATE modifies existing data like student email.
3. DELETE removes specific records such as enrollments.
4. Validates operations using SELECT to ensure correctness.

Task Descriptor- 4: (Aggregate Queries)

INPUT:

```
Database: MySQL v5.7
Run Update Fork Load Example Star PRO

Query SQL PRO
1 -- Count students per course
2 SELECT c.course_name, COUNT(e.student_id) AS student_count
3 FROM Courses c
4 JOIN Enrollments e ON c.course_id = e.course_id
5 GROUP BY c.course_name;
6
7 -- Courses with more than 1 student
8 SELECT c.course_name
9 FROM Courses c
10 JOIN Enrollments e ON c.course_id = e.course_id
11 GROUP BY c.course_name
12 HAVING COUNT(e.student_id) > 1;
```

Query successfully executed in 50.44ms

OUTPUT:

Database: MySQL v5.7

Run Update Fork Load Example Star PRO

Results PRO Copy as Markdown

Query #1 Execution time: 0.2ms

course_name	student_count
There are no results to be displayed.	

Query #2 Execution time: 0.16ms

course_name
There are no results to be displayed.

Explain on:

1. Uses functions like COUNT() to summarize data.
2. GROUP BY groups records based on course names.
3. HAVING filters grouped data (e.g., courses with >1 student).
4. Helps in analysis like enrollment trends per course.

Task Descriptor -5: (Query Optimization)

INPUT:

Database: MySQL v5.7

Run Update Fork Load Example Star PRO

Query SQL PRO

Query successfully executed in 8.19ms

```

1 -- Original
2 SELECT * FROM Students
3 WHERE student_id IN (
4     SELECT student_id FROM Enrollments WHERE course_id = 101
5 );
6
7 -- Optimized
8 SELECT s.*
9 FROM Students s
10 JOIN Enrollments e ON s.student_id = e.student_id
11 WHERE e.course_id = 101;
  
```

OUTPUT:

The screenshot shows a web-based SQL editor interface. At the top, there's a blue header bar with a database icon, the text 'Database: MySQL v5.7', and buttons for 'Run', 'Update', 'Fork', 'Load Example', 'Star', and a 'PRO' badge. Below the header, the 'Results' section is visible, also featuring a 'PRO' badge and a 'Copy as Markdown' button. Two queries are listed: 'Query #1' with an execution time of 0.25ms and 'Query #2' with an execution time of 0.14ms. Each query result is shown in a table with three columns: 'student_id', 'name', and 'email'. Both tables are empty, with the text 'There are no results to be displayed.' below each one.

student_id	name	email
------------	------	-------

There are no results to be displayed.

student_id	name	email
------------	------	-------

There are no results to be displayed.

Explana on:

1. Identifies inefficient queries (e.g., subqueries).
2. Rewrites queries using JOINS for better performance.
3. Ensures both original and optimized queries give same result.
4. Improves execution speed, especially for large datasets.